

Machine Learning Approaches: Simple ML vs. Deep Learning

Workshop Report

1. Selected Examples and Real-World Applications

Simple Machine Learning: Email Spam Detection

Algorithm: Logistic Regression / Decision Tree

Real-world application: Gmail's spam filter. Google processes over 15 billion unwanted messages every day, and its models flag spam and phishing emails with roughly 99.9% accuracy. Gmail combines lightweight, feature-based classifiers with signals such as sender and domain reputation, authentication checks (SPF, DKIM, DMARC), and content cues like flagged keywords or suspicious links. It also continuously retrain on real-time user behavior — for example, when recipients mark a message as spam, delete it unread, or unsubscribe immediately, that feedback becomes new training data.

Deep Learning: Image Recognition (Facial Recognition)

Algorithm: Convolutional Neural Network (CNN)

Real-world application: Facebook's DeepFace system, launched in 2014. DeepFace uses a deep CNN trained on millions of facial images to power automatic photo-tagging across Facebook's platform. Lower layers of the network learn simple patterns like edges and textures, while deeper layers learn complex facial structures such as the position of eyes and the shape of the nose — all learned automatically from raw pixel data rather than hand-coded rules. The system also applies a 3D face-alignment step so it can recognize faces despite differences in pose, angle, and lighting, and it reportedly reached about 97.35% accuracy at a scale of millions of faces.

2. Why Each Approach Fits — and Why the Other Doesn't

Email Spam Detection: Why Simple ML Is Suitable

Spam detection relies on a manageable set of well-understood, engineered features: word frequency, sender reputation, presence of links, header metadata, and similar structured signals. Logistic regression and decision trees are well suited to this kind of tabular, lower-dimensional data. They train quickly, are cheap to run at massive scale (billions of emails per day), and are relatively interpretable, which matters for a system that needs to explain and continuously adjust why a message was flagged. Because the relationship between these features and the spam/not-spam label is fairly direct, a simple model can achieve very high accuracy without the added complexity of deep learning.

Why deep learning is not the better fit here: a CNN or other deep network is built to learn its own features from raw, unstructured, high-dimensional input, such as pixels or raw sequences of text. In spam detection, the useful features are already known and easy to extract, so a deep network would mostly be re-learning what simple engineering already provides — at a much higher computational and data cost, with slower inference, less interpretability, and greater risk of overfitting relative to the size and structure of the available data. At Gmail's scale, the added compute and latency of a full deep model for every message is unnecessary overhead.

Facial Recognition: Why Deep Learning Is Suitable

Facial recognition works with raw image pixels, which are extremely high-dimensional and vary enormously with lighting, angle, pose, and expression. There is no simple, fixed set of hand-crafted features (like "eye shape" or "nose width") that reliably captures a face across all these variations. CNNs are designed exactly for this problem: their layered structure automatically learns a hierarchy of features, starting with simple edges and textures and building up to complex, discriminative facial patterns, directly from the pixel data. This is what allows DeepFace to reach high accuracy across millions of faces in real-world, uncontrolled photo conditions.

Why simple ML is not the better fit here: logistic regression or a decision tree would require someone to manually engineer the relevant facial features in advance, which is impractical given how much faces vary with pose and lighting, and those models cannot capture the complex, non-linear spatial relationships between pixels that a CNN learns automatically. Even with heavy manual feature engineering, a simple model's accuracy on raw image data would fall far short of what a deep network achieves, making it unsuitable for a task like large-scale facial recognition.

Sources

Gmail spam-filtering statistics and process: Google/Gmail machine learning spam filtering coverage (2026).

DeepFace architecture, training scale, and accuracy: Facebook DeepFace (2014) facial recognition system overview.